Problem Definition

Emergence is widely described as the spontaneous appearance of ordered, adaptive-like behavior in complex systems, yet its underlying mechanism remains undefined. Existing accounts rely on descriptive metaphors or domain-specific heuristics without identifying a general causal structure. This paper addresses the foundational question: *What produces emergent order*, and how can its apparent intelligence be formally characterized across substrates?

Proposed Contribution

We propose a reductionist framework in which emergence arises mechanically from a dual-plane temporal architecture—continuous micro-dynamics coupled to a discrete genomic clock—operating under strict constraint geometry. This pseudo-orthogonal temporal structure forces survival-biased pruning to act as a geometric filter, producing RUM and TUM as the dominant long-clock update geometries. The model generalizes evolutionary theory by abstracting selection, inheritance, and fidelity into a universal constraint-driven mechanism that synthetically generates pre-agentive intelligence without invoking agency or domain-specific biology.

Key Contribution

This framework reverses the conventional interpretation of emergence. Chaos is treated not as an exploratory regime but as a perfectly fragile, perfectly balanced static object that cannot survive contact with real constraints. Ordered behavior therefore arises not from the accumulation of complexity, but from the systematic elimination of high-entropy trajectories across deep time. By embedding pruning within a dual-plane temporal architecture, the model identifies the geometric filtration that yields structured, pre-agentive intelligence and unifies evolutionary and artificial systems under the same constraint-driven mechanism.

Theoretical Foundations

The framework formalizes five primitives: a heritable manifold G, accumulated stress S, universal constraints C, a viability kernel V, and a bounded moveset M. A dual-plane temporal model separates continuous micro-dynamics from discrete macro-ticks, allowing pruning to operate as a survival filter. The resulting macro-trajectory minimizes a resistance functional coupling update cost and fragility. Pre-agentive intelligence emerges as a synthetic byproduct of this minimization, reflected in directional momentum, responsiveness, and coherence preservation.

Cross-Domain Mapping

The proposed geometry aligns with cross-disciplinary constructs including constraint topology, alignment under uncertainty, drift-plus-jump dynamics, structural inference, and macro-to-micro coupling. It provides a unifying substrate for interpreting diverse systems—biological, cultural, symbolic, and artificial—through the same primitives, explaining why similar temporal signatures appear across heterogeneous domains.

Scope and Intent

This paper provides a theoretical foundation for emergence and pre-agentive intelligence as products of constraint geometry and deep-time pruning. It does not offer empirical calibration or domain-specific modeling. Its purpose is to define a transferable structure for future analysis and for the companion papers that develop its implications.

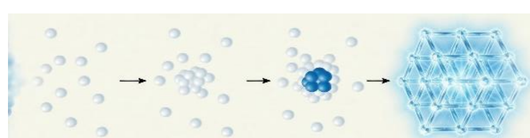

Figure 1: crystallization as the collapse from $O(n!)$ **->** $½^{\boldsymbol{n}}$

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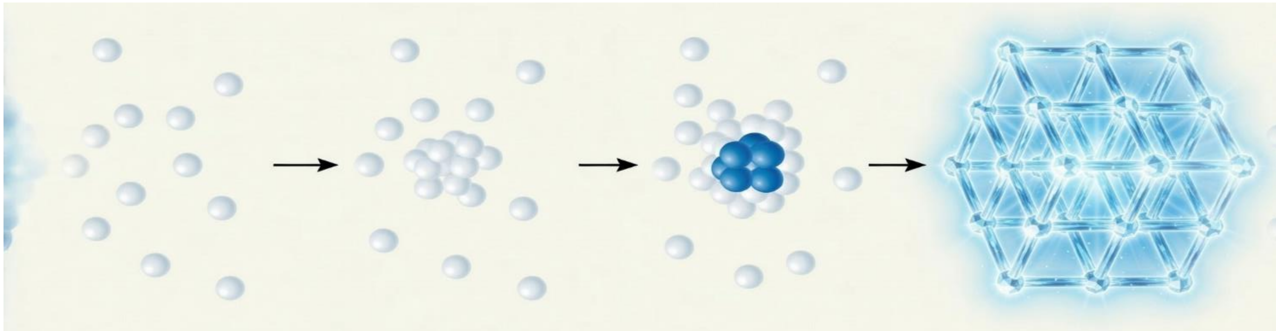

Figure 1. Crystallization as the collapse from $O(n!)$ to $½^n$.
This schematic illustrates why chaos is not a physical regime but a perfectly fragile, static abstraction. The combinatorial space $O(n!)$ exists only under the impossible assumption of zero motion, zero interaction, and zero temporal evolution. The moment a single atom moves, the system undergoes a titration cascade in which unstable configurations are eliminated by bond geometry, energy minima, and constraint coupling. Nucleation marks the first viable point inside the viability kernel, after which deep-time pruning drives the collapse into a low-resistance lattice. Order emerges not by accumulating complexity but by the catastrophic reduction of incompatible states. This same mechanism underlies the synthetic generation of pre-agentive intelligence in the model. (See *Appendix G* for full detail on the perfect fragility of randomness)

Orientation for Interpretation
This paper presents a reductionist framework in which emergence is explained through constraint geometry, dual-plane temporal structure, and survival-biased pruning. It treats chaos not as a generative regime but as a perfectly fragile, static configuration that collapses when subjected to real transmission, fidelity, or energetic constraints. Within this architecture, the genomic or macro-tick clock operates orthogonally to micro-dynamics, enabling pruning to function as the dominant geometric operator over deep time. The resulting filtered trajectories yield structured, pre-agentive intelligence without invoking intention, optimization, or agency. The goal of this work is conceptual: to define a transferable set of primitives and temporal mechanics that unify evolutionary dynamics, information transmission, and artificial systems under a single mechanism. The companion papers provide evidence frames demonstrating how these primitives manifest across biological, cognitive, cultural, and synthetic domains.

Keywords:
constraint geometry, dual-plane time, viability kernel, bounded moves, survival-biased pruning, deep-time collapse, perfect fragility of randomness, high-entropy elimination, temporal attractors, RUM/TUM geometries, synthetic pre-agentive intelligence, emergent structure

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Abstract

Consensus holds that in chaotic, high-dimensional systems, outcomes are dictated by accidental instantiation: random events produce unique results governed by O(n!) combinatorial explosion. This paper challenges that view and finds the opposite.

Chaos is not a stable regime — it is fragile and undergoes entropic decay toward structure in open, dissipative channels. In any system that must transmit coherent information across generations under real constraints (noise, bounded resources, fidelity requirements), high-entropy states (O(n!) combinatorial explosion) are high-resistance and decay rapidly: they cannot maintain signal integrity over deep time. The accessible state space does not expand indefinitely — it shrinks. Randomness is not preserved; it is eliminated through relentless pruning. The living world is empirical proof: what persists is not ever-growing chaos, but increasingly constrained, self-organizing attractors that dissipate energy efficiently while exporting entropy.

Viewed from a point many billions of years after inception, what we call “intelligent” systems crystallize not as products of genius or randomness, but as the inevitable residue of exactly two dominant sorting strategies: Regular Incremental Flow (RUM) — smooth, continuous small adjustments — and Threshold-Triggered Reconfiguration (TUM) — long high-fidelity plateaus punctuated by bounded, coherence-preserving jumps.

The unifying breakthrough is the recognition that organismal dynamics and hereditary propagation occupy pseudo-orthogonal temporal planes: continuous micro-time versus discrete macro-ticks at which the heritable manifold is updated. Between macro-ticks the slow layer remains in high-fidelity copy. This separation enforces mechanical necessity: randomness is pruned, constraints demand embedded error control, and only RUM and TUM geometries (or their mixtures) survive as low-resistance attractors.

Apparent intelligence — the directional, structured behavior observed in any persistent system — is therefore the geometric inevitability of self-structuring under deep-time constraint. This framework does not address biological or agentive intelligence, which requires internally initiated loops and subjective experience; it explains only the residue that looks intelligent once randomness has been eliminated.

What survives is not genius. It is inevitability.

Keywords: survival bias, constraint geometry, pseudo-orthogonal time, temporal attractors, apparent intelligence, self-structuring systems, RUM/TUM geometries

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1) Introduction

Purpose. This paper presents a general, information-theoretic framework for persistent hereditary information under constraint. Five independent domain-specific inquiries—high-stakes relationships, threat processing under ambiguity, cognitive linguistic plasticity (CLP), pre-literate symbolic transmission, and structural decipherment of lost oral cultures—served as empirical probes. Each used biology or psychology as a microscope; the primitive identified here lives below them in the pure geometry of constrained transmission through time.

The Two Systems (kept simple).
We identify two dominant update geometries:

- **RUM — Regular Incremental Flow.** At each macro-event (transmission or reproduction), the heritable object undergoes a small constraint-aligned update. Macro-trajectories are smooth with frequent, low-magnitude adjustments. Average fragility is higher than TUM but peak fragility remains low through continual release.
- **TUM — Threshold-Triggered Reconfiguration.** Between macro-events, the heritable object is held in high-fidelity copy while accumulated stress grows. When stress crosses a viability threshold, the system executes a single bounded reconfiguration that preserves coherence. Macro-trajectories show plateaus punctuated by bounded jumps.

Unifying breakthrough: genomic time as a pseudo-orthogonal plane.
The core insight that unlocked the entire lattice is that the heritable layer updates at discrete macro-ticks, whereas organismal or carrier dynamics evolve in continuous micro-time. These are not two speeds of the same process; they are pseudo-orthogonal temporal planes. Once separated, the rest follows with mechanical necessity:

1. Systems that persist are not products of randomness; deep pruning under constraint preserves only low-resistance update geometries.
2. Real channels with bounded resources require error control (redundancy / coding) to maintain fidelity over long horizons.
3. Under these pressures, exactly two update geometries dominate: RUM (regular incremental release) and TUM (threshold-gated reconfiguration). Mixed regimes can occur, but these two are the attractor poles.

Substrate-agnostic scope; biology/psychology as evidence, not target.
We present an information-theoretic model. Biology and psychology supply illustrative evidence frames only. The same primitives appear in symbolic transmission, cultural memory, and, by natural extension, AI training dynamics where weights (the heritable object) evolve under constraints.

What this is not.
This is not a theory of consciousness, agency, or will. Systems without internal loop initiation can still

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look intelligent because the only long-clock survivors are those that land on directional, coherence-preserving geometries.

Qualification on "Two Systems."
"Two Systems" denotes the dominant survivors. Many other update modes exist (e.g., fallback self-copy, compatibility workarounds, contextual switches). They are non-dominant channels: short-horizon, high-decay, or environmentally brittle. Modeling every transient would destroy tractability without changing the deep result.

How the prior domain probes are used here.
Each domain's findings can be read as an independent validation of the same primitive:

- **Relationships:** bidirectional coupling between fast and slow layers; downward flow of constraints.
- **Threat processing:** bounded-time controller that forces termination via minimal explanatory insertion at thresholds (fast-layer analog of TUM).
- **Linguistic plasticity:** critical-period filter that installs slow-layer structure and defines the viable moveset.
- **Pre-literate symbolic transmission:** collapse to orthogonal error-correcting geometries under extreme memory and noise limits.
- **Lost-culture decipherment:** operational use of the attractors to recover structure from lossy channels.

The details of those studies are not required to read this paper; they are evidence frames that align with—and motivated—this general model.

---

2) Model Primitives

We formalize a minimal, substrate-agnostic model of temporal update under constraint. The goal is explicit testability and transportability across domains.

2.1 Heritable state $\boldsymbol{G}$

Let G_t denote the heritable distribution over configurations on a measurable space $(\mathcal{G}, d)$, where d is a metric used to quantify displacement between successive macro-states. For concreteness in empirical work, d can be chosen as Wasserstein-1 (earth-mover distance) or another topology-appropriate metric. We call G_t the genomic manifold (terminological convenience); what matters is that G_t summarizes the population-level heritable layer at macro-tick t. Between macro-ticks, G is held in high-fidelity copy in the slow layer.

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2.2 Accumulated stress $\boldsymbol{S}$

Let $S_t \in \mathbb{R}_{\geq 0}$ represent accumulated deviation from viability at macro-tick t. Between macro-ticks (in continuous sweep), S integrates small mismatches: mutational load, environmental mismatch, robustness shortfall, coding loss, or analogous terms appropriate to the substrate. At macro-events, part of S is released (RUM: incrementally; TUM: in a bounded jump). If S exceeds a viability threshold τ without release, the lineage decays (extinction) or undergoes forced reconfiguration (TUM).

Operational instantiations. Depending on domain, S can be measured as, e.g., expected fitness deficit, expected fragility under perturbations, or coding reliability deficit under channel noise. The model requires that two investigators given the same data could estimate S consistently.

2.3 Constraints $\boldsymbol{C}$

C encapsulates universal limits: fidelity, energy/resource budget, memory/processing capacity, plasticity windows (temporal opportunity for large changes), and transmission cost/noise. C is fixed for the analysis window (piecewise-constant in practice). Constraints funnel possibilities, biasing toward low-resistance moves.

2.4 Viability kernel $\boldsymbol{V}$

$V \subset \mathcal{G}$ is the set of viable heritable states under constraints C (states that can persist through at least one more macro-interval with non-negligible probability). Intuitively, V encodes what can survive given the channel limits.

2.5 Bounded moves $\boldsymbol{M}$

At each macro-event, the system selects a feasible update $m \in M(G_t, C)$ with $G_{t+1} = m(G_t)$ such that $G_{t+1} \in V$.

- **Hard bounds** exclude updates that violate C outright.
- **Soft bounds** penalize updates that are possible but high-resistance (costly, low probability).

Because M is small relative to $\mathcal{G}$, the process avoids combinatorial explosion—this is why the system can appear “smart” without intelligence: the constraints have already eliminated bad options.

2.6 Dual-phase clock (canonical time)

Time is represented as a dual-phase mechanism:

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- **Continuous sweep (micro-time):** carrier dynamics evolve continuously; S increases via $\dot{S} = \delta(G_t, C, E)$, where E captures environmental parameters.
- **Macro event (discrete, intra-tick):** the heritable layer updates when a state-dependent trigger fires *within* the sweep.

Event rules.

- **RUM (cadenced):** macro events at a regular cadence (or within a narrow jitter band determined by C); at each event, a small move in M.
- **TUM (threshold-gated):** macro event occurs when S crosses τ; a bounded reconfiguration in MMM releases accumulated stress.

Equivalence note (intuition vs formalism).
The dual-phase clock unifies the earlier "two clocks" intuition: a single continuous reference clock with state-dependent discrete events provides the same power while avoiding contradictions. One can view the process as Markovian on the augmented state (G, S) with state-dependent hazard for the next macro-event (equivalently: semi-Markov / PDMP-style). The connection to the Markov foundation is what matters; the label is secondary.

2.7 Assumptions (minimal)

1. **Bounded resources:** constraints C are stationary (or piecewise-stationary) over the analysis window.
2. **Copy-then-update:** between macro-events, the heritable layer is copied at high fidelity; structural updates enter only at macro-events.
3. **Monotone stress:** S is non-decreasing between events and partially releases at events.
4. **Coherence-preserving updates:** all $m \in M$ yield $G_{t+1} \in V$.
5. **No agency:** the process has no internal loop initiation; apparent direction arises from external filtering by constraints and survival bias.

2.8 Glossary of symbols (for standalone clarity)

- G_t: heritable distribution ("genomic manifold") at macro-tick t
- S_t: accumulated deviation from viability at t
- C: universal constraints (fidelity, energy, memory, plasticity windows, transmission cost/noise)
- V: viability kernel (subset of $\mathcal{G}$ that can persist under C)
- $M(G_t, C)$: bounded set of feasible updates
- $\delta(\cdot)$: stress accumulation rate in continuous sweep
- τ: viability threshold on S for TUM triggering
- $d(\cdot,\cdot)$: metric on $\mathcal{G}$ to measure macro displacement

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(Compatibility kernels κ and fallback behavior will be introduced in §4; referenced here only for completeness.)

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3) How the Two Dominant Geometries Emerge

We now explain why RUM and TUM dominate under long-clock survival with constraints.

3.1 Intuition: survival bias + constraint funnels

In high-dimensional configuration spaces, most paths are high-resistance given C: they dissipate energy, violate fidelity, exceed memory, miss plasticity windows, or fail under noise. Over deep time, survival bias eliminates these paths. What remains is a tiny set of low-resistance temporal geometries—the ones that can recycle structure reliably through macro-events. Two qualitatively different strategies survive:

- RUM: frequent, cheap variance release
- TUM: expensive change deferred, then released as a bounded jump

This dichotomy is temporal and geometric, not biological. Biology is merely an instance. These geometries are properties of constrained update systems, not mechanisms of biological reproduction. Biology merely instantiates them.

3.2 RUM (Regular Incremental Flow)

Mechanism. Macro-events arrive on a regular cadence (or within a narrow band). At each event, the system executes a small coherence-preserving update from $M(G_t, C)$. Stress S continually releases, preventing large peaks.

Geometry. Macro-trajectories of G show smooth, wavy paths: $\mathbb{E}[d(G_{t+1}, G_t)]$ is small and frequent.

Fragility profile. Higher average fragility (because the system lives near local adjustments), but lower peaks due to continuous variance release.

Cost-fit. RUM is favored when small changes are inexpensive and the channel supports frequent reliable updates.

3.3 TUM (Threshold-Triggered Reconfiguration)

Mechanism. Between macro-events, G is held in high-fidelity copy while S accumulates. When $S \geq \tau$, the system executes a bounded reconfiguration from $M(G_t, C)$ that resets much of the accumulated stress.

Geometry. Macro-trajectories show plateaus with bounded jumps (“sawtooth”). Between jumps, $\mathbb{E}[d(G_t + \Delta, G_t)]$ is near zero; at a jump, d is bounded by feasibility under C.

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Fragility profile. Lower average fragility during plateaus; higher peaks at jumps.
Cost-fit. TUM is favored when changes are expensive, plasticity windows are rare, or fidelity constraints penalize frequent small updates.

3.4 Mixed regimes and toggling

Real systems often toggle: they run RUM-like within stable bands and switch to TUM-like when stress growth outpaces incremental release. Because both regimes are defined by the same primitives (G, S, C, V, M), toggling requires no new machinery—only parameter changes (e.g., an increase in δ, a temporary reduction in cadence reliability, or a narrowing of plasticity windows).

3.5 Why other modes are non-dominant in the long run

Local deviations (e.g., fallback self-copy, compatibility workarounds, context-specific hybridizations) can be adaptive in the short term, but they tend to be high-decay under steady constraints or collapse into RUM/TUM mixture dynamics. Modeling all such transients would be biologically faithful but mathematically intractable and unnecessary for the long-clock result. The dominant survivors are defined by when stress is released (continually vs at thresholds) and how updates are bounded by C.

3.6 Minimal propositions (testable signatures)

- **Proposition 1 (RUM displacement bound)**. Under fixed C and cadence bounded by $[\bar{\Delta}, \bar{\Delta}]$, there exists $\epsilon(C) > 0$ such that
  $\mathbb{E}[d(G_{t+1}, G_t) \mid RUM] \leq \epsilon$ and
  $Var(d(G_{t+1}, G_t) \mid RUM)$ is small.
  Interpretation: frequent, small adjustments produce wavy macro-signatures with low peak stress.
- **Proposition 2 (TUM jump bound)**. Under fixed C and viability threshold τ, there exist bounds $0 < \epsilon < \Delta(C, \tau)$ such that between jumps, $\mathbb{E}[d(G_t + \Delta, G_t)] \leq \epsilon$ and at a threshold event, $d(G_{t+}, G_{t-}) \leq \Delta$.
  Interpretation: long plateaus punctuated by bounded reconfigurations yield sawtooth macro-signatures with higher peak stress.
- **Proposition 3 (fragility trade-off)**. For identical C and comparable long-run throughput, RUM exhibits lower peak stress and higher average stress than TUM; TUM exhibits the reverse.
  Interpretation: peak vs average fragility separates regimes.

(Formal proofs and simulation sketches appear in the Appendix; they rely on standard drift-plus-jump arguments and hazard controls for event timing.)

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4) Compatibility & Fallback

We now formalize partner compatibility and the fallback mechanism, both of which are required to keep the model realistic yet mathematically tractable.

4.1 Compatibility kernel κ

At a macro-event that involves merging or recombination, let each candidate carrier contribute a heritable sample $g_i, g_j \in \mathcal{G}$ drawn from the current population (or, in non-biological settings, from the active archive). Define a compatibility kernel

$$\kappa(g_i, g_j \mid C) \in [0,1]$$

as the probability that merging (g_i, g_j) produces a coherence-preserving update $G_{t+1} \in V$ under constraints C. The kernel implicitly encodes biochemical constraints, regulatory constraints, coding rules, or any substrate-specific compatibility condition; we only require that κ be bounded and measurable. The expected merge success at macro-tick t is

$$R_t \;=\; \mathbb{E}_{(g_i,g_j)\sim G_t \times G_t}[\kappa(g_i, g_j \mid C)].$$

When a merge succeeds, an incremental update (RUM) or bounded reconfiguration (TUM) is drawn from $M(G_t, C)$ and a fraction of stress S_t is released. When a merge fails, no structural update is recorded in the slow layer and stress continues to accumulate.

Interpretation: κ is a generalized error-correcting filter: it disallows merges that would violate C or produce incoherent offspring/outputs. It is also how non-random pairing, assortativity, spatial structure, and mate choice can be represented without proliferating new modes.

4.2 Binary complementarity (polarity) as a special case

A widely useful special case is binary complementarity (“polarity”). Each carrier has a label $P \in \{+1, -1\}$. Only opposite labels are compatible:

$$\kappa_{pol}(g_i, g_j) \;=\; 1\{P_i \neq P_j\}.$$

If labels are unstructured in the population (random pairing; no assortativity), let p_t be the proportion of $+$ at tick t. Then the effective mixing rate is

$$R_t \;=\; 2\, p_t\, (1 - p_t),$$

maximized at $p_t = 0.5$ This is Lemma 1 (binary complementarity, random pairing) and should be read strictly under its assumptions. It’s useful because it quantifies how balance accelerates mixing and how skew slows it; but we treat it as one kernel among many, not a universal law.

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4.3 Fallback Principle (continuity under scarcity)

Long-clock persistence requires that the macro-event not stall indefinitely. We therefore posit a Fallback Principle:

If a system fails to secure a compatible merge within a bounded number of opportunities or within a bounded window inside the dual-phase sweep (as set by C), it reverts to a continuity-preserving update:
– copy/self (TUM-like), or
– deferred update with minimal release (RUM with near-zero step).

Formally, let $W(C)$ be the maximum allowed waiting time (or maximum number of failed attempts) before continuity must be asserted. If no (g_i, g_j) yields a successful κ-pass in $W(C)$, then $G_{t+1} = G_t$ (copy) or $G_{t+1} \in M(G_t, C)$ with a minimal step. Either route prevents extinction by deadlock while respecting C.

Why we include this: It preserves mathematical functionality and recovers observed fallbacks (e.g., self-copy, selfing, sex-change, apomixis, or code-path bypass) without declaring new "systems." These are non-dominant channels that keep time moving when compatible partners are scarce.

4.4 Stress accounting with compatibility

We couple κ to stress:

- Successful merge → $S_{t+1} = S_t - r_{succ}(G_t, C)$.
- Failed or skipped merge → $S_{t+1} = S_t + \delta(G_t, C, E)$ (accumulation).
- Fallback copy → small change to G; S increases unless a bounded reconfiguration is performed (TUM event).

These rules ensure that rarity of compatible partners is reflected as higher stress loading, which in turn shifts event timing (earlier TUM triggers; altered cadence reliability) in the dual-phase clock.

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5) Geometric Signatures & Metrics

This section defines observable, testable differences between RUM and TUM. The goal is to make the theory empirically falsifiable and portable.

5.1 Metric on the heritable space

Let $d(\cdot,\cdot)$ be a metric on $\mathcal{G}$ (e.g., Wasserstein-1). Define the step series

$$\Delta t \;=\; d(G_t + 1, G_t).$$

We also track stress peaks $max\ S$ and stress averages $\bar{S}$ over observation windows.

5.2 Canonical RUM vs TUM signatures

With identical C, RUM and TUM show distinct macro-patterns:

- RUM (wavy):
  $E[\Delta_t]$ small; $Var(\Delta_t)$ small to moderate; few large outliers.
  Fragility: higher average stress ($\bar{S}$) but lower peaks ($maxS$) due to constant release.
- TUM (sawtooth):
  Long stretches with $\Delta t \approx 0$ punctuated by bounded spikes; $Skew(\Delta_t)$ strongly positive.
  Fragility: lower average stress but higher peaks at jumps.

We formalize these as inequalities in the Appendix (Propositions 1–3). In short: peaks separate TUM; averages separate RUM.

5.3 Event hazards in the dual-phase clock

Let $H(\cdot)$denote the hazard for firing a macro-event inside the continuous sweep.

- RUM: cadence-like hazard, approximately constant within the viability band (or narrowly distributed around a schedule set by C).
- TUM: threshold hazard increasing in S; near zero far from τ, rising steeply as $S \to \tau$.

This produces distinct inter-event distributions: tighter for RUM, heavy-tailed (or at least over-dispersed) for TUM.

5.4 Identification protocol (practical)

Given an event series and states $\{G_t\}$:

1. Compute step metrics Δt, their empirical mean/variance/skew, and peak/average stress (or proxies).
2. Fit a simple hazard model: constant vs S-sensitive.
3. Classify the regime by:

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    - peak/average stress trade-off (RUM: low peaks/high averages; TUM: high peaks/low averages),
    - dispersion of inter-event times (RUM: narrow; TUM: broader),
    - skewness of steps (RUM: low; TUM: high).
4. Test stability under subsampling and window shifts; mixed regimes will show segment-wise changes in these diagnostics.

5.5 Fragility bands and viability margins

Define a fragility band $\mathcal{F}_\alpha$ as the α -percentile band of S over an observation window. RUM tends to keep $\mathcal{F}_\alpha$ narrow and centered higher, TUM keeps it wider and lower but with upper outliers at jumps. Viability margins (distance to τ) behave accordingly: RUM reduces variance of margin; TUM improves average margin but tolerates near-breach episodes.

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6) Apparent Intelligence as Residue

We now articulate exactly what we mean by "appears intelligent."
Claim. In any system where a heritable object must persist under constraints, survival bias prunes the vast configuration space down to low-resistance update geometries. The two dominant survivors (RUM, TUM) generate directional, structured behavior because incoherent directions have been eliminated by the filter. Over deep time, the surviving trajectories show:

- Directionality (momentum concentrates along feasible moves M),
- Responsiveness (stress modulates event timing and move selection),
- Robustness (updates are coherence-preserving with respect to V), and
- Problem-solving appearance (progress on hard constraints looks like "insight," but is the residue of pruning).

No agency is required. The system does not decide to be intelligent; it cannot avoid appearing intelligent once the long-clock filter has removed everything else.

Boundary with actual intelligence. Biological or artificial agency requires internally triggered loops (willful initiation of fast-layer dynamics that change the slow-layer prospects). This paper does not attempt to model agency, consciousness, or subjective experience. It shows why non-agent systems look agentive whenever the only survivors are those that ride RUM/TUM geometries under C.

> Optional note for readers who connect this to AI: If a training or selection process imposes a long-clock filter on model parameters (weights as the heritable object), the resulting dynamics can exhibit RUM/TUM signatures and produce outputs that appear purposive. This follows from the same primitive; no claim about machine "thought" is implied.

6.1 Definition of Apparent Intelligence

In this framework, apparent intelligence is a purely mathematical property of the long-term trajectory of the heritable manifold ($G(t)$):

> **Apparent intelligence** is the observable tendency of (G(t)) to exhibit
> (i) directional momentum along low-resistance paths in the constrained configuration space,
> (ii) responsiveness of update timing and magnitude to accumulated stress (S(t)), and
> (iii) robustness of signal integrity (coherence with respect to the viability kernel (V)) under repeated transmission through noisy, resource-bounded channels.

Formally, a system displays apparent intelligence to the degree that its macro-trajectory minimizes the total resistance

$$R = \int \left(c\left(G, \dot{G}\right) + af(G)\right)dt$$

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subject to the constraint manifold (C), where (c) is the local update cost and α weights robustness. The degree can be quantified inversely to the surviving combinatorial volume after K macro-ticks, which collapses exponentially (e.g., roughly $1/2^K$ during early pruning phases). The two geometries RUM and TUM are the global minima of this functional under the joint pressures of fidelity, noise, and intergenerational transmission.

Apparent intelligence is therefore not an emergent "achievement" but a geometric inevitability: it is what remains after all higher-resistance (more chaotic, higher-entropy) trajectories have been pruned by survival bias across deep time.

This definition is deliberately narrow and mathematical. *(To the author's knowledge, this is the first time emergence has been formally defined as the degree of minimization of a resistance functional under deep-time pruning, with exponential growth in survived macro-ticks.)* It makes no reference to agency, consciousness, subjective experience, or biological intelligence, all of which require internally initiated control loops and are explicitly outside the scope of this paper.

6.2 Emergence

Emergence—conventionally: spontaneous, ordered, adaptive-like behavior in complex systems—is treated here as a temporal property of trajectories, not of instantaneous states. It accrues along $G(t)$ as the count of survived macro-ticks increases. Each macro-tick applies a survival filter: only updates that preserve signal integrity (coherence with respect to V) and keep fragility $f(G)$ below catastrophic levels can propagate. Let $p_k \in (0,1)$ be the conditional survival probability at tick k. The expected surviving combinatorial volume after K ticks is

$$V_{survived}(K) \approx V_{initial} \times \prod_{\{k=1\}}^{K} p_k$$

so that each $p_k < 1$ acts as a multiplicative contraction of accessible phase space. Under deep-time dynamics, p_k begins small (harsh pruning) and tends toward 1 as the trajectory enters an attractor basin; early phases therefore collapse accessible chaos rapidly, followed by asymptotic stabilization on low-resistance geometries.

Define apparent intelligence inversely to remaining accessible chaos:

$$I(K) \propto \frac{1}{V_{survived}(K)} \propto exp\left(-\sum_{\{k=1\}}^{K} \log p_k\right)$$

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When p_k is small and roughly constant, growth in $I(K)$ is approximately exponential in the number of pruning events; as $pk \to 1$, growth slows with stabilization. Directional momentum, stress-responsive timing, and robustness increase only because high-resistance branches have already been eliminated.

Observer frame and "simulation"

Natural dynamics do not define problems, goals, or boundaries. They do not specify a start, an end, a population, or a metric. Patterns become intelligible only after an observer frame $\mathcal{F}$:

$$\mathcal{F} = (\mathcal{T}_0, \mathcal{T}_1,\ \mathcal{U},\ \mathcal{M})$$

where $[\mathcal{T}_0, \mathcal{T}_1]$ bounds time, $\mathcal{U}$ specifies the comparison set (e.g., trajectories or states), and $\mathcal{M}$ is the metric/criterion used to evaluate persistence or structure. Relative to $\mathcal{F}$, pruning resembles problem-solving: unstable configurations are eliminated; stable ones persist; persistence appears "optimal" only because $\mathcal{F}$ segments and measures it. In this strict sense, emergence simulates intelligence: it matches the formal shape of a simulation (bounded interval, population, and metric), even though the underlying dynamics include none of these.

Variational form (inevitability under constraints)
Given the resistance functional from §6.1,

$$\min_{G(\cdot)} \int_0^T \left[c\left(G(t), \dot{G}(t)\right) + \alpha\, f\big(G(t)\big) \right] dt$$

subject to $G(t) \in V$, stress accumulation $\frac{dS}{dt} = \lambda + \xi(t)$, hazard-gated event timing $h(S)$, bounded update kernel $\Delta \sim M(G, C)$, and intergenerational fidelity embedded in C, the global minima are the two temporal geometries RUM and TUM (and mixtures under nonstationary C). They are not contingent surprises but the forced signatures of deep-time pruning under survival bias. Apparent intelligence, when perceived through $\mathcal{F}$, is the observer-relative reading of this minimization, not a system property

Emergence simulates intelligence only because observers impose the boundaries of a simulation: a beginning, an end, a population, and a metric. The underlying system contains none of these. The appearance of intelligent organization arises from the imposed frame, not from the intrinsic dynamics.

Science and mathematics are engineered tools for producing replicable observations. They work by constructing simplified, repeatable conditions around processes that are not intrinsically segmented or repeatable. When these tools are applied to natural or artificial systems, the imposed frame defines what counts as structure, coherence, or intelligence.

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In this sense, artificial systems—including AI—inherit the intelligence attributed to them from the engineered conditions of their training and evaluation. The intelligence is not a natural property; it is a property of the measurement frame. The system exhibits structured behavior because the constraints, segmentation, and replicability conditions enforce it. This is artificial emergence: structured behavior that becomes *intelligible* only relative to the imposed frame, not from any intrinsic problem-definition in the system itself.

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7) Evidence Frames (Illustrative, not exhaustive)

Each frame briefly shows how a domain probe aligned with the primitives. These are independent validations, not prerequisites.

- Relationships (compatibility & constraint flow). Interaction patterns reflected a fast/slow coupling: immediate choices (fast layer) were gated by slow-layer constraints, consistent with the dual-phase clock. The compatibility kernel metaphor operationalized partner fit without invoking teleology.
- Threat processing (bounded-time controller). Under sustained ambiguity, systems demonstrated thresholded termination via minimal explanatory insertion — a cognitive analog of TUM where internal stress (load) accumulates until a minimal, coherence-preserving reconfiguration resolves the state.
- Linguistic plasticity (critical-period filter). Early windows installed structure into the slow layer, defining parts of C and narrowing M. This is a direct instance of plasticity windows as a universal constraint that shapes the viable moveset.
- Pre-literate symbolic transmission (error control primitives). With tiny working memory and noisy acoustic channels, redundancy constrained by fidelity and bandwidth collapsed the feasible signaling space to a small number of orthogonal ECC geometries, demonstrating that constraints require error control.
- Lost-culture decipherment (operational attractors). The same primitives were used as decoding scaffolds to recover structure from low-SNR cultural remnants, consistent with viability kernels and bounded moves under severe C.
- Physical analogies (demonstrations only). Sand dunes (RUM-like stable flux) and glaciers (TUM-like plateau + calving) illustrate how similar temporal geometries arise whenever update costs and plasticity windows differ substantially.

Each domain agrees with the same primitives: (G, S, C, V, M), dual-phase clock, compatibility kernel κ, and fragility trade-offs — reinforcing that the lattice is information-theoretic rather than substrate-specific.

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8) Predictions & Falsifiers

To keep the framework accountable, we list specific predictions and ways to disconfirm them.

8.1 Predictions

1. Peak/average trade-off. Under identical C, RUM regimes exhibit lower peak S and higher average $\bar{S}$ than TUM; TUM exhibits the reverse.
2. Inter-event dispersion. RUM shows narrow inter-event distributions; TUM shows over-dispersion consistent with state-dependent hazards.
3. Step skewness. The Δ_t distribution is low-skew under RUM and high-skew under TUM.
4. Compatibility scarcity shifts timing. Reductions in $R_t = \mathbb{E}[\kappa]$ advance threshold events (TUM) or stretch cadence windows (RUM), observable as timing drifts in the dual-phase clock.
5. Fallback observables. Episodes of partner scarcity yield copy/self or near-zero steps, measurably increasing S and biasing toward TUM-like behavior until compatibility recovers.
6. Mixed regime markers. Under non-stationary C, diagnostics switch segment-wise (cadence → threshold or vice versa) with predictable changes in peaks/averages.

8.2 Falsifiers

- Counter-trade-off: Observing RUM with higher peaks than TUM under matched C and equivalent throughput would contradict the model.
- Timing invariance to scarcity: If large, sustained decreases in R_t do not shift event timing or stress profiles, the κ coupling is wrong.
- No signature separation: If Δ_t skew and inter-event dispersion fail to separate regimes across controlled simulations, the geometry claims are overstated.
- Inoperable fallback: If continuity persists without compatible merges and without fallback, the dual-phase timing or stress logic is incomplete.

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9) Limits & Non-Claims

- **No agency.** We do not model consciousness, will, or subjective experience.
- **Dominance**, not exclusivity. "Two Systems" denotes dominant survivors; other modes exist but do not persist as primitives.
- **Constraint sensitivity.** Results depend on the correct specification of CCC; mis-specifying fidelity, bandwidth, plasticity windows, or energy budgets will distort signatures.
- **Short-horizon deviations**. Over brief intervals or during shocks, observations can look nothing like their long-clock geometry; this is expected and not diagnostic by itself.
- **Model reduction**. We intentionally omit domain-specific complexities that would make the model intractable without improving the deep explanatory value.

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10) Figure Set

1. **Dual-phase clock equivalence.** Left: intuition (fast vs slow planes). Right: formal single sweep with state-dependent intra-tick event and stress-weighted hazard.
2. **State space with viability kernel**. V shaded; arrows show feasible $M(G, C)$; hard/soft bounds annotated.
3. **Hazard curves**. RUM: cadence-like hazard (flat/narrow). TUM: threshold hazard rising with S (sharply increasing near τ.
4. **Trajectory signatures**. Overlay Δ_t series and S for matched C. Panels: RUM (wavy, low peaks) vs TUM (sawtooth, high peaks).
5. **Compatibility kernel**. Heatmap of $\kappa(g_i, g_j)$; inset: binary complementarity with $R = 2p(1-p)$ labeled Lemma 1 (assumptions: random pairing, binary kernel).
6. **Fragility bands**. RUM vs TUM distributions of $\bar{S}$, $max\ \ S$, and margin to τ (viability).

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Appendix A: Process Class (Formal Skeleton)

- **State**: $X_t = (G_t, S_t) \in \mathcal{P}(\mathcal{G}) \times \mathbb{R} \geq 0$
- **Between events (continuous sweep):** $\dot{S} = \delta(G_t, C, E)$; G held in copy mode.
- **Event hazard**: $h(X) = h(G, S; C)$.
- **Event draw**: waiting time $T \sim Exp(\int h)$ or a renewal with state-dependent intra-tick.
- **Update at event**: pick $m \in M(G, C)$; set $G^+ = m(G), S^+ = S - r(G, C)$ (bounded release); or if $TUM \ \wedge \ S \geq \tau$, draw a bounded reconfiguration with larger r.
- **Compatibility:** success with probability $R_t = \mathbb{E}[\kappa]$; else fallback per §4.
- **Classification**: RUM if $h \ \approx$ cadence-like and $\mathbb{E}[\Delta_t]$ is small/frequent; TUM if h is dominated by S-threshold crossings.

This can be realized as a semi-Markov or PDMP-style process; proofs only require boundedness and measurability assumptions.

A.1 Mathematical Core

The model is a piecewise-deterministic Markov process on the augmented state
$X(t) \ = \ (G(t), S(t))$

where

- $G(t) \ \in \ \mathcal{g}$ is the heritable manifold (a probability distribution or point in configuration space with metric d, e.g., Wasserstein-1),
- $S(t) \ \geq \ 0$ is accumulated informational stress (scalar or low-dim vector measuring deviation from viability).

Continuous sweep (between macro-events):

$$\frac{dG}{dt} \ = \ 0 \text{ (high-fidelity copy mode)}$$
$$\frac{dS}{dt} = \lambda \ + \ \xi(t)$$

where

- $\lambda \ > \ 0$ is baseline stress accumulation rate
- $\xi(t)$ is weak noise (e.g., Gaussian or bounded).

Stress accumulates monotonically until an event fires.

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A1.2 Macro-event hazard function:

Events occur according to a state-dependent hazard h(S):

- RUM regime:

$$h(S) \approx \text{constant}$$

  or narrow cadence band, e.g., periodic with small jitter).
- TUM regime:

$$h(S) = \begin{cases} 0, & S < \tau \\ H_{sharp}(S), & S \geq \tau \end{cases}$$

  where h_{sharp} increases rapidly as $S \to \tau$ (step-like or exponential).

This represents cadenced vs threshold-gated macro-event timing.

A1.3 Transition kernel at macro-event (update rule):

At an event time t_k, an update is drawn from the bounded moveset:

$$\Delta \sim M(G(t_k^-), C)$$

subject to universal constraints C (fidelity ε, energy/memory budget B, noise σ, plasticity window δ).

Then:

$$G(t_k^+) = G(t_k^-) + \Delta$$

Stress is released according to:

$$S(t_k^+) = \begin{cases} S(t_k^-)\,(1 - \rho_{small}), & RUM\ event \\ S(tk -)\,(1 - \rho large), & TUM\ jump \end{cases}$$

A 1.4 Viability kernel:

Let:

$$V \subset \mathcal{G}$$

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be the compact set of viable hereditary states under constraints C.
Define fragility:

$$f(G) = dist(G, \partial V),$$

the distance of G to the viability boundary.

A1.5 Regime classification & toggling conditions:

A long-run trajectory is:

- **RUM-dominant** if inter-event times cluster around a cadence $T_{cadence}$ and

$$\mathbb{E}[d(G+, G)] \ is \ small.$$

- **TUM-dominant** if inter-event times are right-skewed and

$$occasional \ bounded \ jumps \ occur.$$

Toggle condition (mixed regimes):
The system shifts regime when the marginal cost of maintaining the current mode exceeds that of the alternative. Example:

$$\mathbb{E}[f(G)]RUM > \theta \quad and \quad max \ f(G)TUM \ is \ tolerable \ \Rightarrow \ shift \ toward \ TUM.$$

A 1.6 Key propositions (sketched consequences):

- **RUM:**
  - bounded per-event displacement
  - low peak fragility
  - higher average fragility

- **TUM:**
  - zero displacement between events
  - bounded jumps at thresholds
  - lower average fragility
  - higher peak fragility

- **Matched long-run throughput:**
  The two regimes exhibit **opposite** trade-offs in average vs peak fragility (Proposition 3).

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A.1.7 Markov property

All dynamics are Markovian on the augmented state (G,S):

- Drift and hazard depend only on current (G, S).
- Event times are drawn from a state-dependent hazard.
- Updates are drawn from a bounded kernel $M(G, C)$.

The process is well-posed under:

- boundedness of M
- Lipschitz continuity of $f(G)$ on V
- finite hazard growth near thresholds

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Appendix B: Propositions (Sketches)

The following statements formalize key consequences of the model. Full proofs follow standard techniques in piecewise-deterministic Markov processes (PDMP), renewal theory, and constrained optimization; they are omitted here for brevity but can be supplied on request.

Prop 1 (RUM displacement bound)
Under a narrow cadence band and bounded moveset M, there exists $K < \infty$ such that the per-event displacement $d\left(G_{\{t+1\}}, G_t\right) \leq K$ almost surely, and the variance of displacements remains small over long horizons.
(Sketch: Cadence enforces frequent sampling from M; bounded radii + stress drain limit step size accumulation.)

Prop 2 (TUM jump bound)
With compact viability kernel V and bounded reconfiguration radius r, off-event periods satisfy $d(G(t)$, $G(t + \Delta t)) = 0$ (copy mode), and on-event jumps satisfy $d\left(G_{\{t+1\}}, G_t\right) \leq r$. Plateaus maintain coherence until threshold crossing.
(Sketch: High-fidelity copy freezes G between events; reconfiguration is constrained by $M(G, C)$.)

Prop 3 (Fragility tradeoff)
For regimes matched on long-run throughput (total displacement over horizon T), average fragility $\mathcal{E}$[f(t)] is higher in RUM than TUM, while peak fragility $max_t f(t)$ is lower in RUM than TUM.

(Sketch: RUM releases stress/variance frequently (low peaks, higher average drift); TUM stores then sheds in bursts (high peaks at jumps, lower average during plateaus). Follows from drift-plus-jump decomposition and hazard monotonicity in S.)

Lemma 1 (Binary complementarity under random pairing)
Under random pairing and a compatibility kernel κ with maximum at complementary states (+/− polarity), the expected mixing efficiency is maximized at $p = ½$ with value $2p(1 - p) \rightarrow ½$.

(Sketch: Standard quadratic optimization over mixing probability under symmetric kernel; reduces to classic Hardy–Weinberg-like form in limit.)

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Appendix C: Minimal Simulation Protocols

Goal. Demonstrate that, under identical constraint parameters (C), the two dominant geometries produce distinguishable signatures in trajectory shape, inter-event statistics, fragility profiles, and stress dynamics — providing existence proofs for the claims in §3 and Propositions 1–3.

All simulations are abstract and low-dimensional: the heritable state G is represented as a point or low-dim vector in $\mathbb{R}^n$ ($n \leq 5$ typical), with a simple metric d (e.g., Euclidean or Wasserstein-1 approximation). No domain-specific biology, psychology, or symbolics is required.

Common setup (matched constraints across regimes)

- Choose fixed constraints C: fidelity bound ε, energy/memory budget B, transmission noise σ, plasticity window δ.
- Define viability kernel $V \subset \mathbb{R}^n$ as a compact convex set (e.g., ball or box).
- Define bounded moveset M(G, C): set of feasible Δ such that $G + \Delta \in V$ and $||\Delta|| \leq$ radius r (hard/soft bounds).
- Stress accumulation rate λ (continuous sweep).
- Stress release fraction ρ at events ($0 < \rho \leq 1$; larger for TUM jumps).
- Threshold τ for TUM triggering.
- Metric d on state space for displacement measurement.
- Run for long horizon T (>> typical inter-event time) with multiple independent realizations.

RUM protocol (Regular Incremental Flow)

- Macro-events occur on fixed cadence schedule (period $T_{cadence} \pm$ small jitter band J).
- At each event:
  - Draw small move $\Delta \sim M(G, C)$ (uniform or gradient-biased toward lower fragility).
  - Update $G \leftarrow G + \Delta$.
  - Release stress: $S \leftarrow S \times (1 - \rho_{small})\ or\ S \leftarrow max(0, S - \delta S)$.
- Record: inter-event times (near-constant), displacements $d(G_{\{t+1\}}, G_t)$, stress trajectory $S(t)$, fragility $f(t) = dist(G(t), \partial V)$.

TUM protocol (Threshold-Triggered Reconfiguration)

- Between events: G held fixed (high-fidelity copy), S integrates as $dS/dt = \lambda + noise$.
- Event triggers when $S \geq \tau$.
- At event:
  - Draw bounded reconfiguration $\Delta \sim M(G, C)$ with larger possible radius (but still bounded).
  - Update $G \leftarrow G + \Delta$.
  - Large stress reset: $S \leftarrow S \times (1 - \rho_{large})$ or $S \leftarrow S_{reset}$ (small residual).
- Record: inter-event times (variable, right-skewed), large occasional displacements, stress sawtooth pattern, fragility drops post-jump.

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Compatibility & fallback extension (optional but recommended)

- Define kernel $\kappa(G1, G2) \in [0,1]$ (e.g., graded or binary polarity).
- At merge opportunity: success with prob κ → viable mixed G'; failure → fallback to copy/self (TUM-like stasis).
- Induce scarcity by setting average κ low; observe fallback episodes as long plateaus.

Outputs & checks (required for each run)

1. Trajectory plots: $G(t)$ projected $(1-2\ dims)\ +\ overlaid\ S(t)$ → confirm wavy (RUM) vs sawtooth (TUM).
2. Displacement histogram: small/frequent (RUM) vs rare/large (TUM).
3. Inter-event time distribution: narrow around cadence (RUM) vs exponential-like or threshold-shaped (TUM).
4. Fragility bands: compute time-averaged fragility $\mathcal{E}[f]$ and peak fragility $max[f]$; confirm Prop 3 tradeoff (higher average but lower peaks in RUM; reverse in TUM).
5. Hazard curves: empirical hazard rate of events vs accumulated S → flat/narrow for RUM, sharply increasing near τ for TUM.
6. Falsifiability checks:
   - Under matched throughput (similar total displacement over long T), verify Prop 1 (RUM bounded small steps), Prop 2 (TUM bounded jumps + plateaus), Prop 3 (opposite average/peak fragility).
   - If signatures do not separate cleanly under identical C, constraints were insufficiently binding or moveset too permissive.

Implementation notes

- Use any standard numerical environment (Python/NumPy/SciPy, Julia, Matlab, etc.).
- Keep n small; avoid high-dimensional chaos.
- Add weak stochasticity to stress accumulation and move selection to prevent degenerate cases.
- Code should be short (~100–200 lines per regime) and deterministic given seed.

These protocols are intentionally minimal: their purpose is only to reproduce the qualitative geometric signatures and quantitative tradeoffs predicted by the model under matched constraints. They are not calibrated reproductions of any real biological/cultural system.

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Appendix D: Origins of the Primitives – Independent Domain Probes

The primitives formalized in this paper (dual-phase clock, accumulated stress S, viability kernel V, bounded moveset M, error-control necessity, and the pseudo-orthogonality of genomic time) first appeared, in fragmented form, across five self-contained inquiries. Each probe targeted a distinct human system under severe constraints and extracted structural regularities without presupposing the general framework presented here.

These works remain complete and standalone: they require no reference to the present synthesis, can be read or archived independently, and serve here only as illustrative evidence that the same geometric residues emerge when the constraint-and-survival-bias lens is applied to different domains.

- **High-stakes relationships** ("Toward a Structural Model of Relationship Compatibility")
  Examined pre-commitment stability under irreversibility, legal entanglements, and shared assets. Surfaced bidirectional coupling between fast-layer autonomy (emotional vs. logical reasoning modes) and downward constraint flow from the slow layer; reframed compatibility as topological alignment in a ternary behavioral manifold (Parsimony-Leverage-Risk) where fragility measures distance from low-resistance paths.
  doi: "10.5281/zenodo.18050285

- **Threat processing under prolonged ambiguity** ("Hallucinated Agents: A Cognitive Mechanism for Resolving Ambiguous Threats")
  Modeled rapid resolution of multimodal uncertainty to avoid physiological coherence fracture. Revealed a bounded-time controller that enforces termination via minimal explanatory insertions (Minimal Viable Agent externally; Internal Minimal Viable Cause internally) when unresolved time and composite load exceed thresholds — an operational fast-layer analog of threshold-triggered reconfiguration (early form of TUM).
  doi: "10.5281/zenodo.18674477

- **Cognitive linguistic plasticity** ("Cognitive Linguistic Plasticity")
  Mapped languages onto a five-dimensional structural coordinate system (lexeme density, morphological flexibility, syntax depth, orthographic transparency, semantic granularity) that calibrates the supervisory cognitive layer during the critical period. Demonstrated language as a developmental lattice/filter gating what information reaches the slow layer; highlighted how early-installed structure defines default heuristics and viable movesets for abstraction, memory, and generalization.
  doi: "10.5281/zenodo.18805427

- **Pre-literate symbolic transmission** ("Language as Constraint Geometry")
  Derived the collapse of redundancy architectures in channels with acoustic degradation, working-memory bounds (≈4±1 chunks), and zero external storage. Proved exactly three

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pairwise orthogonal, globally attracting geometries (discrete-algebraic lattice, bounded-curvature continuous contour, rigid phase-multiplexed clocks) as the only stable solutions; showed these reduce to mixtures of regular incremental and threshold-triggered update rhythms under fidelity and memory constraints. Extended the same manifold to non-human biological channels (DNA reading frames, cetacean song, pheromone gradients).
doi: "10.5281/zenodo.18807009

- **Structural decipherment of lost oral cultures** ("A Universal Decoder for Lost Oral Cultures") Operationalized the three geometric primitives as error-correcting attractors calibrated during the critical period. Used universal experiential keystones (e.g., dawn/noon/dusk; birth/growth/death) as alignment anchors to recover probabilistic dictionaries from undeciphered scripts/recitations via slot-to-keystone mapping and noise subtraction — direct application of the attractor geometry to inference in bottleneck-constrained transmission.
doi: "10.5281/zenodo.18807013

Each probe independently converged on subsets of the same primitives because the underlying constraint manifold (noisy transmission, bounded memory, intergenerational fidelity, survival bias) admits few long-clock stable geometries. The present work abstracts those recurring elements into a unified, substrate-agnostic model.

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Appendix E: Glossary (Complete)

- G_t: heritable distribution ("genomic manifold") at macro-tick t
- S_t: accumulated deviation from viability at t
- C: constraints (fidelity, energy, memory, plasticity windows, transmission cost/noise)
- V: viability kernel within $\mathcal{G}$
- $M(G_t, C)$: bounded set of feasible, coherence-preserving updates
- δ: stress accumulation rate in continuous sweep
- r: stress release at event (size depends on update type)
- τ: viability threshold for TUM triggering
- κ: compatibility kernel $[0,1]$
- d: metric on $\mathcal{G}$ (e.g., Wasserstein-1)
- Δ_t: step size $d(G_{t+1}, G_t)$
- H, h: event hazard(s) in the dual-phase clock
- R_t: expected merge success (effective mixing rate)
- $\mathcal{F}_\alpha$: fragility band (percentile band of S)

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Appendix F: The Impossibility of Persistent Chaos – Why O(n!) Is Not a Viable Regime

A persistent assumption across many fields is that chaotic, high-dimensional systems are governed by accidental instantiation: random events draw from an effectively infinite O(n!) combinatorial space, producing unique outcomes that can somehow be preserved indefinitely. This view is internally paradoxical.

If the system is truly chaotic, then "preserved indefinitely" should be impossible. Chaos, by definition, erodes correlations, destroys fidelity, and diffuses information. The only apparent defense of the paradox is an appeal to mathematical infinity ("in the abstract, this could continue forever"). Yet even under that defense, the conclusion is the same: chaos is not a neutral or robust default — it is a state of continuous entropic decay toward structure. Randomness does not await ordering; it collapses into ordering the moment any real transmission, coupling, or constraint is allowed to act.

I. Mathematical Potential vs Physical Reality

On paper, a system with n components admits O(n!) possible configurations — an astronomical blank check. In physical reality, the moment any two components can interact, bond, couple, or influence one another, the pure-random regime collapses. Uncontrolled coupling immediately destroys the independence and isotropy that define randomness. The combinatorial explosion is therefore not a stable library of possibilities; it is a transient, unstable cloud that disintegrates as soon as physics is switched on.

II. Randomness as a Terminal, Non-Transmissible State

A truly random state cannot persist across even modest timescales in any system that must transmit coherent information or structure. High-entropy configurations are maximally fragile: they diffuse correlations so broadly that no coherent signal survives noise, resource limits, or fidelity requirements. Any attempt to copy or propagate such a state results in rapid decoherence — the randomness "melts" into irretrievable loss.

Empirical reality confirms this fragility everywhere we look. No persistent system on Earth exhibits unbounded, stable randomness at the relevant scale. If chaos were viable, we would observe ever-expanding combinatorial diversity preserved across deep time. Instead we observe the relentless elimination of high-entropy states.

III. The Environment as a Relentless Filter (Illustrative Example)

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As a concrete illustration, consider written English. The theoretical state space of 26 letters yields 26^n possible strings of length n — an astronomical number. Yet meaningful language emerges and persists precisely because the vast majority of those combinations are “illegal” under the constraints of human physiology, cognition, phonetics, syntax, and cultural transmission. The environment acts as a grammar that collapses the combinatorial explosion before it can manifest as persistent noise. Next-word prediction is possible only because the functional state space is tiny compared to the mathematical one.

The same filtering operates universally. The physical environment — forces, energetic budgets, noise floors, transmission costs — supplies the grammar. Unconstrained randomness is illegal syntax: it cannot be copied twice without catastrophic loss. Persistent chaos is not a default; it is a terminal condition that nature discards immediately.

IV. Conclusion

Persistent randomness is impossible because any real system allows interactions that immediately destroy the independence required for pure chaos. O(n!) describes a mathematical fiction, not a physical regime that can endure. What we observe in nature is not the preservation of combinatorial explosion, but its systematic elimination. Randomness does not await structure — it collapses into structure the moment physics is allowed to act.

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Appendix G: The Perfect Fragility of Randomness – Why O(n!) Cannot Persist

Randomness is widely assumed to represent the "default" or "baseline" state of high-dimensional systems. This appendix formalizes why this assumption fails. We show that true randomness — in the sense required by the combinatorial abstraction $O(n!)$ — is not a physical regime but a perfectly fragile, measure-zero configuration that collapses immediately when any dynamical variable is introduced.

This appendix clarifies:

(i) why chaos cannot persist

(ii) why perfect randomness can exist only as a static mathematical ideal

(iii) why real systems undergo catastrophic collapse toward low-resistance geometries under even infinitesimal perturbations.

G.1 Randomness as a Measure-Zero, Static Abstraction

The combinatorial expression $O(n!)$ often appears in complexity literature as the size of the "accessible" state space. Formally, it enumerates the number of distinct micro-configurations of n elements under complete independence.

However, this enumeration is implicitly atemporal: no interactions, no momentum, no coupling terms, no energy gradients, no constraints, no correlation structure, no update path, no viability requirement.

In this strict sense, $O(n!)$ describes only the cardinality of mathematical possibility, not a space of physically maintainable dynamical states.

Thus, randomness as defined by $O(n!)$ is not a region of state space but a static point floating in the algebra, disconnected from any permitted physical trajectory. Randomness becomes "perfect" — but only in the trivial sense that it exists only when nothing is allowed to move.

G.2 Perfect Fragility: Why Chaos Cannot Survive Motion

A system in an idealized random state is perfectly fragile. This means:

- any motion
- any energy fluctuation
- any collision
- any drift of a single particle
- any binding opportunity
- any local gradient

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instantly destroys the independence assumptions that define “randomness.”

Once independence collapses, the combinatorial space collapses with it.

This collapse is catastrophic because the system transitions from a combinatorial description to a geometrically constrained description. Constraints eliminate the overwhelming majority of configurations in a single step.

One perturbation triggers a titration cascade:
- A single atom moves.
- Local energy minima shift.
- Neighbors adjust to maintain coherence.
- Microstructure forms.
- The initial symmetry of the random cloud is broken.
- Constraint geometry becomes active.
- Non-viable configurations are eliminated.
- The system falls into a pruning regime.

High-entropy possibility collapses into a narrow viability kernel. This is exactly the physical mechanism behind nucleation and crystallization. Randomness does not “evolve.” It is eliminated.

G.3 Catastrophic Collapse as the Fundamental Generator of Structure

Once the system leaves the measure-zero random configuration, it is forced into a cascade of eliminations. Every step removes degrees of freedom that cannot satisfy constraints such as bond geometry, conservation laws, fidelity thresholds, energy minima, memory limits, error tolerance, or bounded movesets.

This collapse is not slow. It is not exploratory. It is instantaneous on the timescale of micro-dynamics. The system moves from “infinite potential” to “strictly bounded viability” in a single transition.

This is the fundamental insight illustrated by the crystallization process: chaos is not a dynamic regime but a rapid-failure mode that instantly transitions into structure.

G.4 Dual-Plane Time as the Collapse Engine

The dual-plane temporal model explains why perfect randomness cannot survive:
- Micro-time introduces perturbations (continuous motion, collisions, diffusion).
- Macro-time applies viability filtering (macro-events, pruning, bounded updates).

Together, they form a two-stroke engine that guarantees collapse:
- Micro-plane: break independence, introduce correlation.

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- Macro-plane: remove incoherent trajectories.

This interaction ensures that no system can remain in or return to the O(n!) abstraction. Once motion exists, randomness is permanently destroyed.

This is why emergent structure is inevitable under deep time: every timestep annihilates chaos further.

G.5 The Rarity of True Randomness (and Why π Is Special)

Only mathematical objects that are:

- defined without dynamics
- defined without coupling
- defined as infinite symbolic expansions

can exhibit perfect, nonrepeating randomness.

Numbers like π are rare not because they are "complicated," but because:

- they exist entirely outside physical constraints
- they operate in a purely symbolic domain
- their randomness is definitionally preserved by isolation from motion, energy, or noise

Physical systems cannot achieve this isolation.
Thus, perfect randomness is common in mathematics but nonexistent in nature. Nature has no mechanism to maintain it.

G.6 Summary

Randomness is physically impossible except as a static, measure-zero snapshot. The slightest motion induces a cascade that collapses combinatorial possibility into narrow geometric constraints. This collapse, driven by dual-plane temporal dynamics and survival-biased pruning, inevitably produces structured trajectories.

Perfect randomness exists only in mathematics. In nature, it is destroyed instantly by time.

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